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Visual Outcomes after Proton Therapy for Recurrent Uveal Melanoma

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Objective: Proton beam re-irradiation (PBI) remains an effective and globe-preserving alternative to enucleation in the treatment of local recurrence in uveal melanoma. The study aimed to assess visual outcomes and prognostic factors in visual acuity (VA) after proton beam salvage therapy.

Design: Retrospective study.

Subjects: A retrospective study evaluated patients with recurrent uveal melanoma treated with PBI from 1984 through 2019 at a single academic tertiary center.

Methods: Patient and tumor characteristics were collected from the medical record, as well as best visual acuity (BVA) and ocular outcomes after treatment of recurrent uveal melanoma with PBI.

Main Outcome Measures: The primary outcome of the study was the BVA of patients after PBI for recurrent uveal melanoma. Additional outcome measures included enucleation rate of patients after salvage PBI and analysis of tumor and patient characteristics in the prognostication of VA.

Results: The study comprised 67 patients who received PBI for recurrent uveal melanoma. The median age at recurrence was 67.6 years (range, 31.6−91.0 years), and median follow-up from the time of recurrence to last examination was 4.4 years (range, 0.23−17.1 years). The median final BVA was hand motions (range, 20/20 to no light perception) and 6 (9.1%) patients maintained a Snellen VA 20/40 or better. The 5-year probability of VA retention of 20/200 or better was 19%. In a multivariable Cox model, VA at tumor recurrence of worse than 20/40 was found to be significantly associated with a VA of 20/200 or worse after retreatment with PBI. Twelve (18%) patients underwent enucleation after retreatment with PBI.

Conclusions: Proton beam irradiation for the treatment of recurrent uveal melanoma allows for ocular preservation and functional vision in select patients.

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Uveal melanoma (UM) is the most common intraocular malignancy in the adult population, with a mean age-adjusted incidence in the United States reported as 5.1 per million.[1] Excellent local tumor control has been achieved with the use of conservation treatment, in particular radiation therapy. The most accepted modalities for globe-retaining radiotherapy include episcleral plaque brachytherapy, heavy charged particles (proton beam therapy, helium ion therapy), gamma knife surgery, and CyberKnife radiotherapy. Local recurrence of UM after radiotherapy ranges from 2% to 10% in the majority of study populations.[2−9] Proton beam irradiation (PBI) has been shown to achieve effective local control, with 2% to 5% of patients experiencing local tumor recurrence requiring additional treatment.[2,8]

The treatment for local recurrence of UM depends on initial tumor management, type of recurrence, tumor characteristics, and patient status. Treatments for recurrent UM typically include enucleation, brachytherapy, proton beam therapy, and transpupillary thermotherapy (TTT). Although secondary enucleation was once regarded as the treatment of choice, re-irradiation of the eye has become a viable alternative for patients seeking globe preservation. Studies have shown local rerecurrence after PBI as salvage therapy ranges from 0% to 30% at 5 years.[10−14] The rates of control after proton beam salvage therapy are comparable to other forms of irradiation after local tumor recurrence.[15−17]

Re-irradiation versus enucleation tends to remain a joint decision between physician and patient. Studies have suggested that overall survival and metastasis-free survival after irradiation for UM recurrence are not compromised by conservative management as compared with enucleation.[11−17] Additionally, patients have a high likelihood for eye retention despite radiation-related ocular morbidity, and a proportion of patients retain functional vision. Few studies have analyzed functional visual outcomes among patients with local UM recurrence treated with PBI. The aim of this study was to evaluate visual outcomes and factors associated with visual prognosis in

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